

MACHINE LEARNING APPROACH TO DETECT MUSCLE FATIGUE

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CHAPTER ONE: INTRODUCTION

BACKGROUND

Muscle fatigue during squat exercises is a common problem that can affect performance.

To address this, a practical solution is to develop a machine learning model that can predict muscle fatigue. Machine learning algorithms analyze various data inputs such as electromyography signals, heart rate, and oxygen consumption to make accurate predictions.

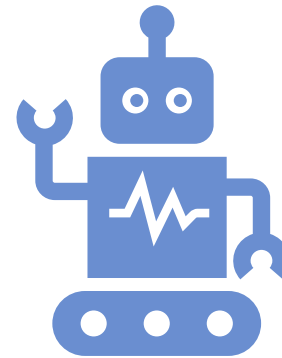
This approach offers objective and personalized feedback to athletes, helping them optimize their training and prevent injuries.

By identifying individuals prone to fatigue-related injuries, these models can also contribute to injury prevention strategies. Machine learning has the potential to enhance sports science and improve training programs, but further research is needed to refine the models and determine the best combination of input variables.

PROBLEM STATEMENT

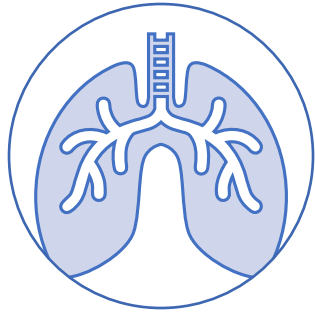


Muscle fatigue during exercise can impact performance and injury risk. Traditional detection methods are not always accurate or reliable. Machine learning algorithms, along with EMG and respiratory sensors, are being explored to accurately detect muscle fatigue.

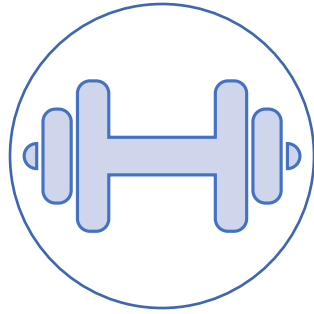


This study focuses on creating a machine learning system that detects fatigue during squats using these sensors. The system will provide real-time analysis, helping athletes and trainers optimize training and prevent injuries.

AIM OF THE PROJECT



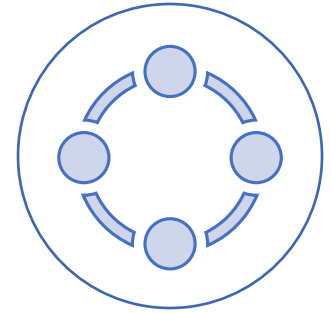
The project aims to develop a machine learning system using EMG and respiratory sensors to accurately detect and monitor muscle fatigue during squat exercises.



This system will provide real-time feedback on fatigue levels, helping optimize workouts and prevent injuries.



It will also allow athletes to understand their physical limits and track progress over time.

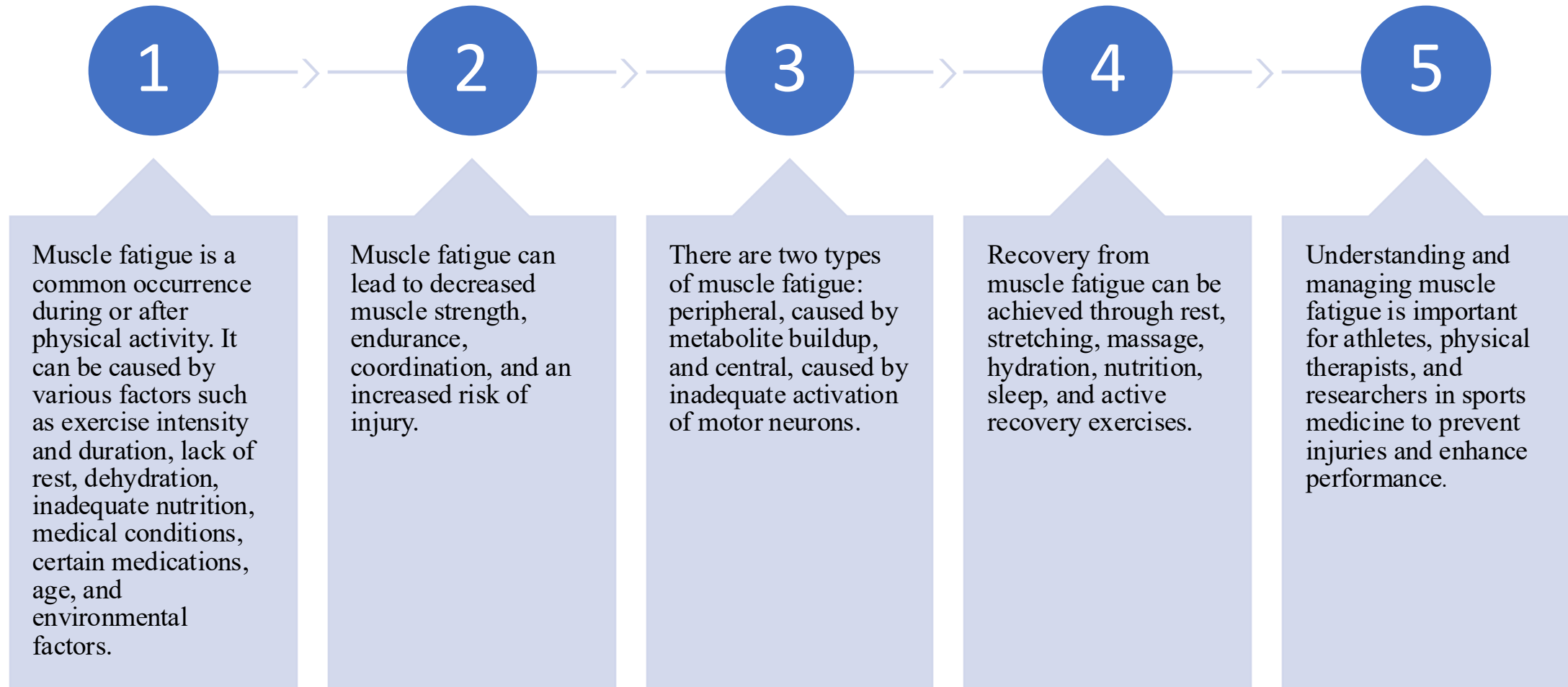


This presentation will discuss everything that concerns muscle fatigue and the machine learning approach



CHAPTER 2 : LITERATURE REVIEW

WHAT IS MUSCLE FATIGUE



EXISTING WORK

STUDY	YEAR	METHODOLOGY	BEST ALGORITHM	ACCURACY
H.B Acharya	2018	sEMG signals, machine learning algorithms	SVM	96.15%
H.B Acharya	2019	sEMG signals, machine learning algorithms	SVM	95.48%
M. S. Sun	2018	sEMG signals, machine learning algorithm	SVM	93.33%
W. Huang, Z	2019	sEMG signals, machine learning algorithm	SVM	94.3.%
J. Lu	2020	sEMG signals, machine learning algorithm	SVM	92.2%

EXISTING WORK

STUDY	YEAR	METHODOLOGY	BEST ALGORITHM	RESULTS
Y. K.Kim	2017	Wearable sensors, machine learning approach	Random forest	94.36%
Y. Song	2020	sEMG signals, machine learning approach	SVM	95.4%
M. D. S Nascimento	2021	sEMG signals, machine learning approach	SVM	96.67%
L. Hu	2020	sEMG signals, deep learning approach	DNN	90.6%
H. Zhang	2019	sEMG signals, machine learning algorithms	SVM	91.4%

EXISTING WORK

STUDY	YEAR	METHODOLOGY	BEST ALGORITHM	ACCURACY
A. K. Al Fahdawi	2020	sEMG signals, ANFIS model	ANFIS	92.5%
Y. Xie	2021	sEMG signals, CNN, LSTM	CNN+LSTM	91.5%
D. Yang	2021	sEMG signals, CNN, LSTM	CNN+LSTM	91.2%
A. D. Nayak	2021	sEMG signals, machine learning algorithm	SVM	92.8%
M. J. Zarei	2021	sEMG signals, CNN, RNN	CNN+RNN	93.5%
D. Han	2021	sEMG signals, machine learning algorithm	SVM	91.7%

SUMMARY

Multiple literature reviews have investigated the use of machine learning techniques to detect muscle fatigue during squat exercise.



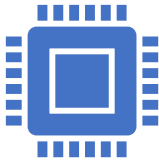
These studies employed various algorithms such as SVM, KNN, decision tree, random forest, CNN, LSTM, and ANFIS, in combination with surface electromyography (sEMG) signals or wearable sensors.



The results consistently demonstrated high accuracies ranging from 90.6% to 96.67% in detecting muscle fatigue levels.



The findings highlight the efficacy of machine learning approaches in real-time monitoring and prediction of muscle fatigue, providing valuable insights for optimizing training routines and minimizing the risk of injuries during squat exercises.



CHAPTER THREE: MATERIALS AND METHODOLOGY

A Vernier instrumentation amplifier is a specialized amplifier used to amplify small signals from EMG sensors.

It has high gain and high input impedance, making it suitable for detecting muscle fatigue during squats.

By connecting the output of EMG sensors to the amplifier, it amplifies the electrical signals generated by muscles, facilitating measurement and analysis.

These amplified signals are then processed by a machine learning system to train a model capable of identifying patterns of muscle fatigue.

The model analyzes frequency or amplitude changes in the signals to detect fatigue and provides real-time feedback during squats.

This feedback can alert users to take breaks or adjust their form to prevent injury or enhance performance.

The Vernier instrumentation amplifier improves the accuracy and reliability of EMG sensors, benefiting athletes and physical therapists in optimizing training and rehabilitation programs.



ELECTROMYOGRAPH

EMG sensors measure the electrical activity produced by muscles during contraction, providing valuable information about muscle behavior.

In a machine learning system, these sensors are attached to specific muscles like the quadriceps or hamstrings during squats to continuously measure muscle activity.

The collected electrical signals are sent to the machine learning system for analysis. Using this data, the system builds a model that predicts muscle fatigue based on patterns of electrical activity.

The model identifies changes in frequency or amplitude that indicate the onset of fatigue.

Once trained, the model offers real-time feedback through a smartphone app or wearable device, alerting users to changes in muscle activity.

This feedback helps users adjust their form or rest, optimizing performance and preventing injury during squats.



SURFACE ELECTROMYOGRAPH (sEMG)

Surface Electromyography (SEMG) is a non-invasive technique that measures and records the electrical activity of muscles using electrodes placed on the skin.

SEMG sensors capture muscle fiber activity during contraction and relaxation, providing insights into force, timing, and coordination.

These sensors find applications in sports and rehabilitation, monitoring muscle activity, assessing fatigue, and aiding in proper muscle use.

In a machine learning system, SEMG data trains models to detect muscle activity patterns and offer real-time feedback during exercises.

By analyzing electrical activity, such as amplitude, frequency, timing, and coordination, the system identifies indicators of muscle fatigue.

This information helps adjust exercise programs, prevent injuries, and optimize performance. SEMG electrodes are typically attached to muscles like quadriceps or hamstrings during squats, picking up signals during contractions.

Machine learning then enables the system to detect and predict muscle fatigue, providing feedback via wearables or smartphone apps. SEMG proves valuable in optimizing training programs, reducing injury risk, and enhancing results.



RESPIRATORY SENSOR (EKG)

A respiratory sensor is a device that measures different aspects of breathing, such as inhalation and exhalation rates, tidal volume, and other respiratory parameters.

When used in a machine learning system to detect muscle fatigue during squats, the sensor is typically placed on the chest or abdomen to track breathing movements

During squats, the sensor continuously measures respiratory activity, and this data is sent to the machine learning system for analysis.

The system combines the respiratory data with information from other sensors, such as SEMG sensors, to build a model that predicts muscle fatigue.

Changes in the user's breathing pattern, such as increased inhalation or exhalation rates, may indicate reaching physical limits.

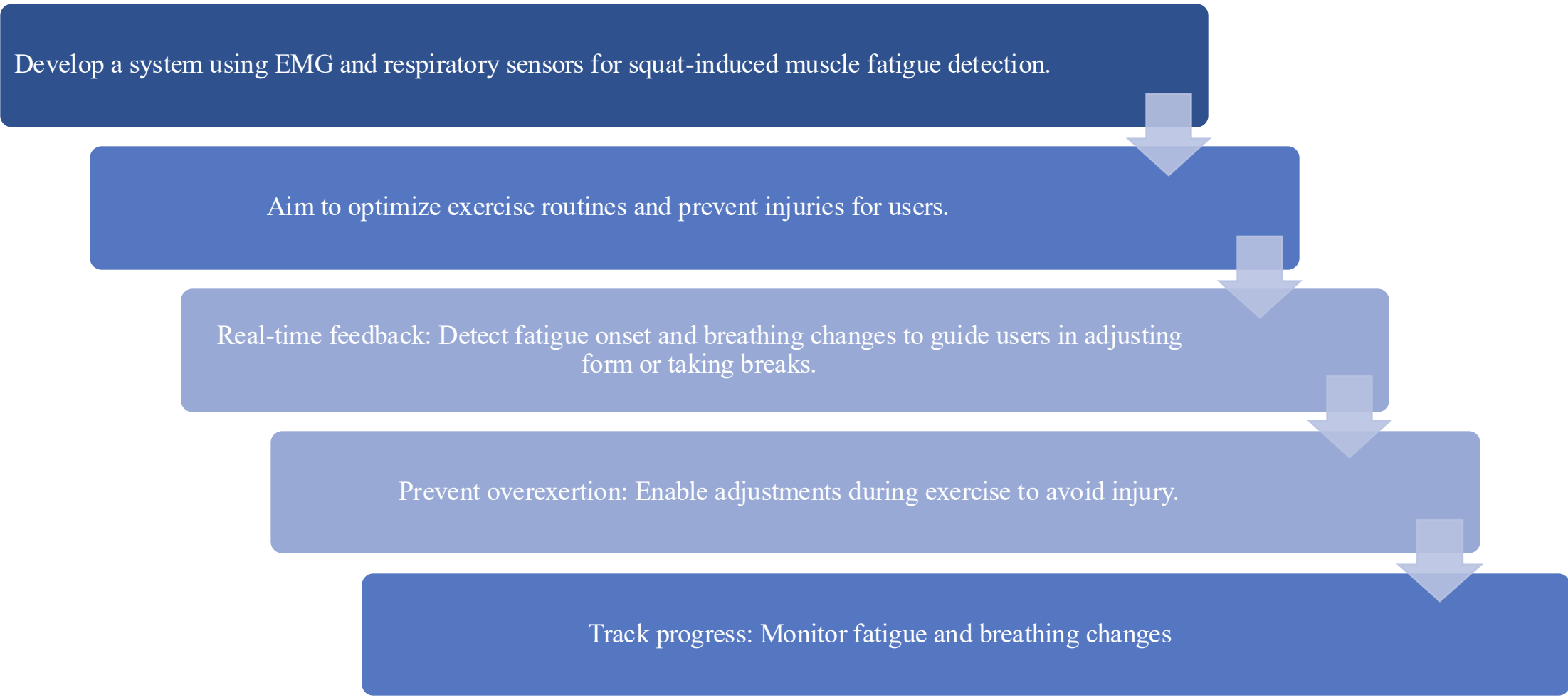
Once the model is trained, it provides real-time feedback to the user during squats, alerting them to fatigue-related breathing changes.

This feedback allows users to adjust their form or take breaks as necessary, optimizing performance and preventing injuries.

In summary, respiratory sensors, integrated with machine learning, offer valuable insights into breathing patterns, enhancing the detection of muscle fatigue during squats and enabling users to optimize their training and minimize injury risks.

OUR GOAL

Develop a system using EMG and respiratory sensors for squat-induced muscle fatigue detection.



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graph TD; A[Develop a system using EMG and respiratory sensors for squat-induced muscle fatigue detection.] --> B[Aim to optimize exercise routines and prevent injuries for users.]; B --> C[Real-time feedback: Detect fatigue onset and breathing changes to guide users in adjusting form or taking breaks.]; C --> D[Prevent overexertion: Enable adjustments during exercise to avoid injury.]; D --> E[Track progress: Monitor fatigue and breathing changes];
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Aim to optimize exercise routines and prevent injuries for users.

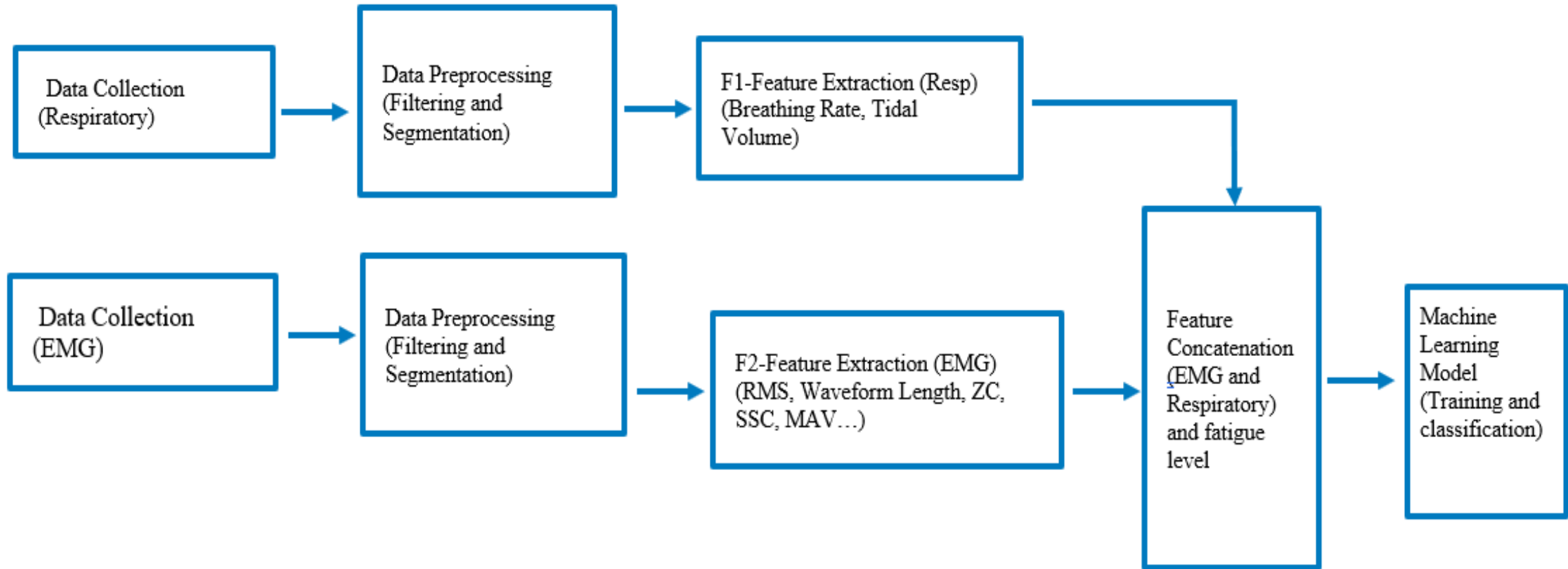
Real-time feedback: Detect fatigue onset and breathing changes to guide users in adjusting form or taking breaks.

Prevent overexertion: Enable adjustments during exercise to avoid injury.

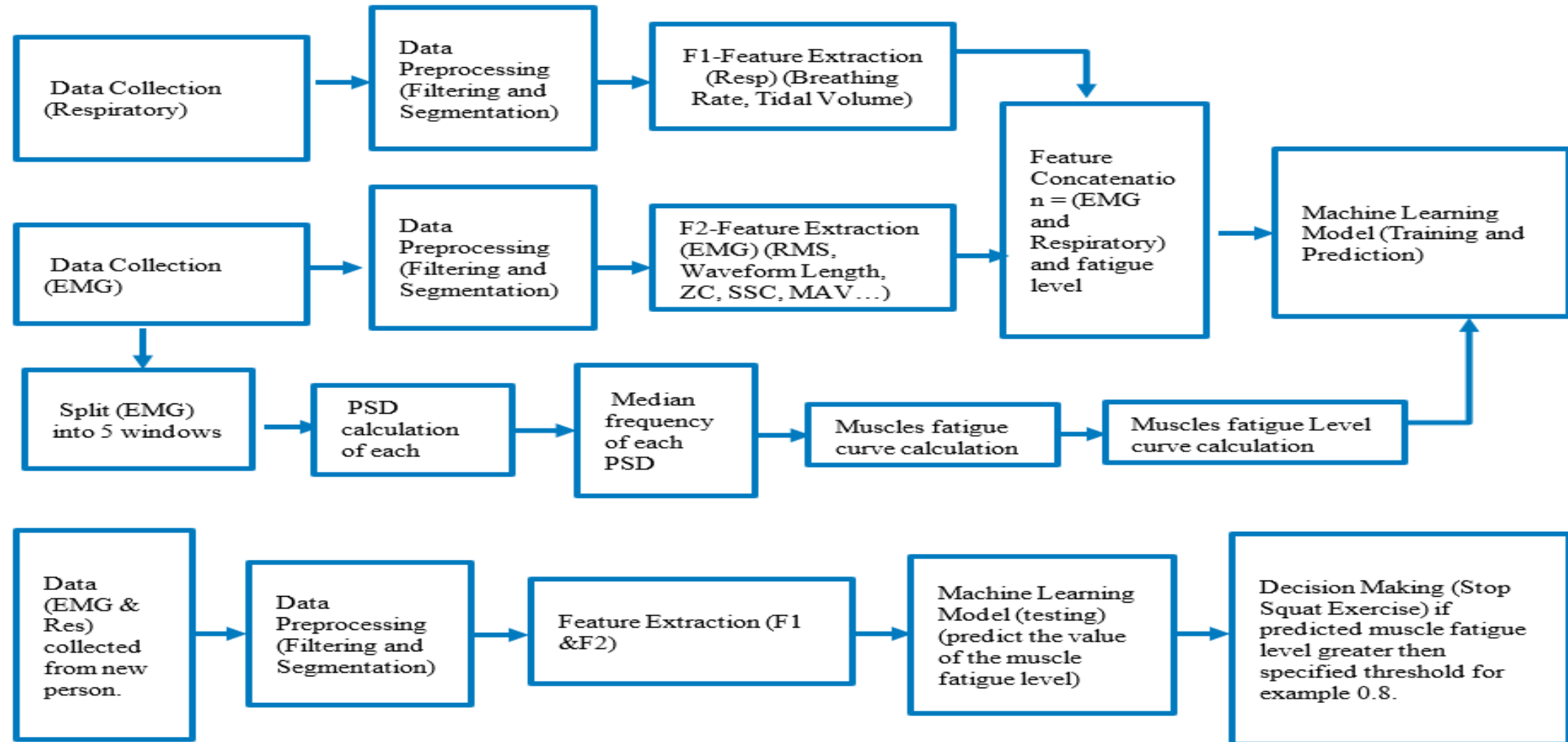
Track progress: Monitor fatigue and breathing changes

METHODOLOGICAL APPROACH

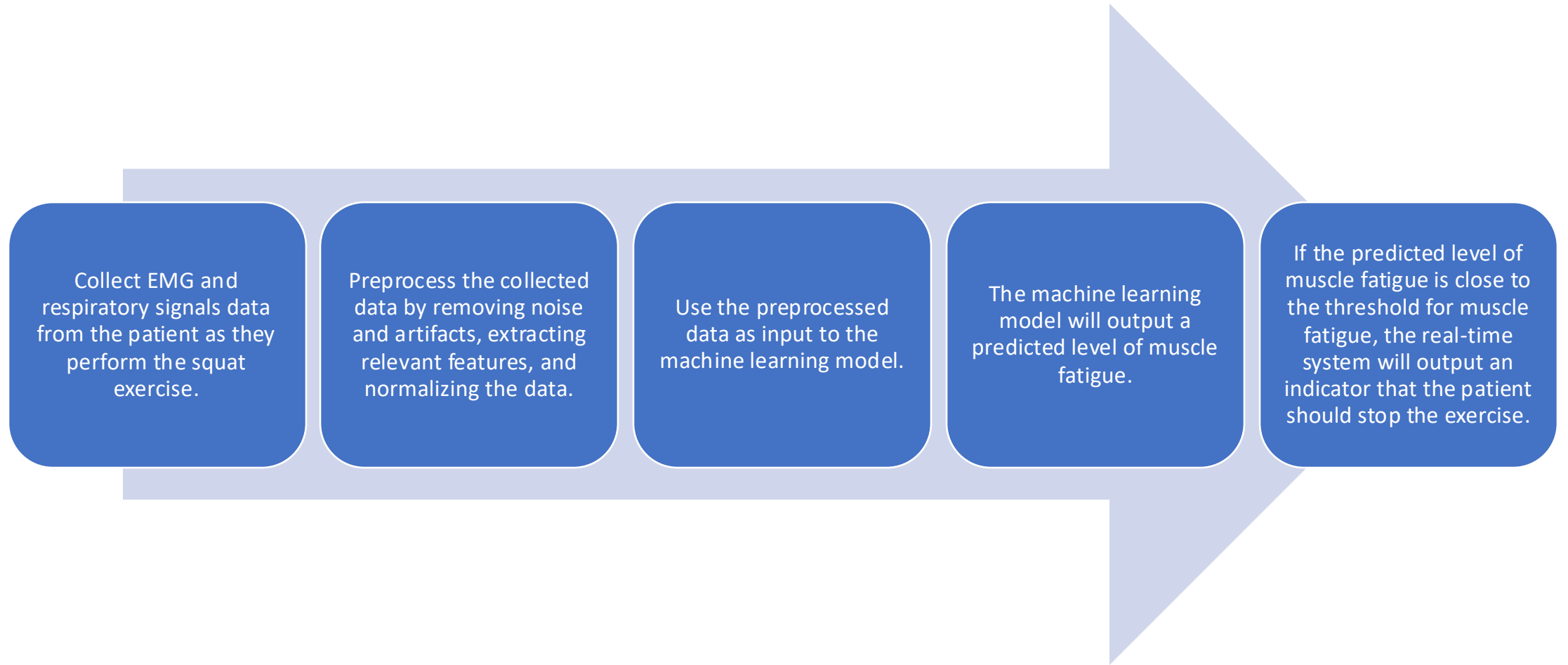
Block diagram discussing the methodology of the work:



FATIGUE LEVEL PREDICTION BLOCK DIAGRAM:



PREDICTION OF FATIGUE LEVEL METHODOLOGY



THE PROPOSED PROGRAMMING METHODOLOGY FOR DEVELOPING THE MACHINE LEARNING SYSTEM IS AS FOLLOWS:

Use MATLAB as the primary programming language for data preprocessing, feature extraction, and machine learning model development.

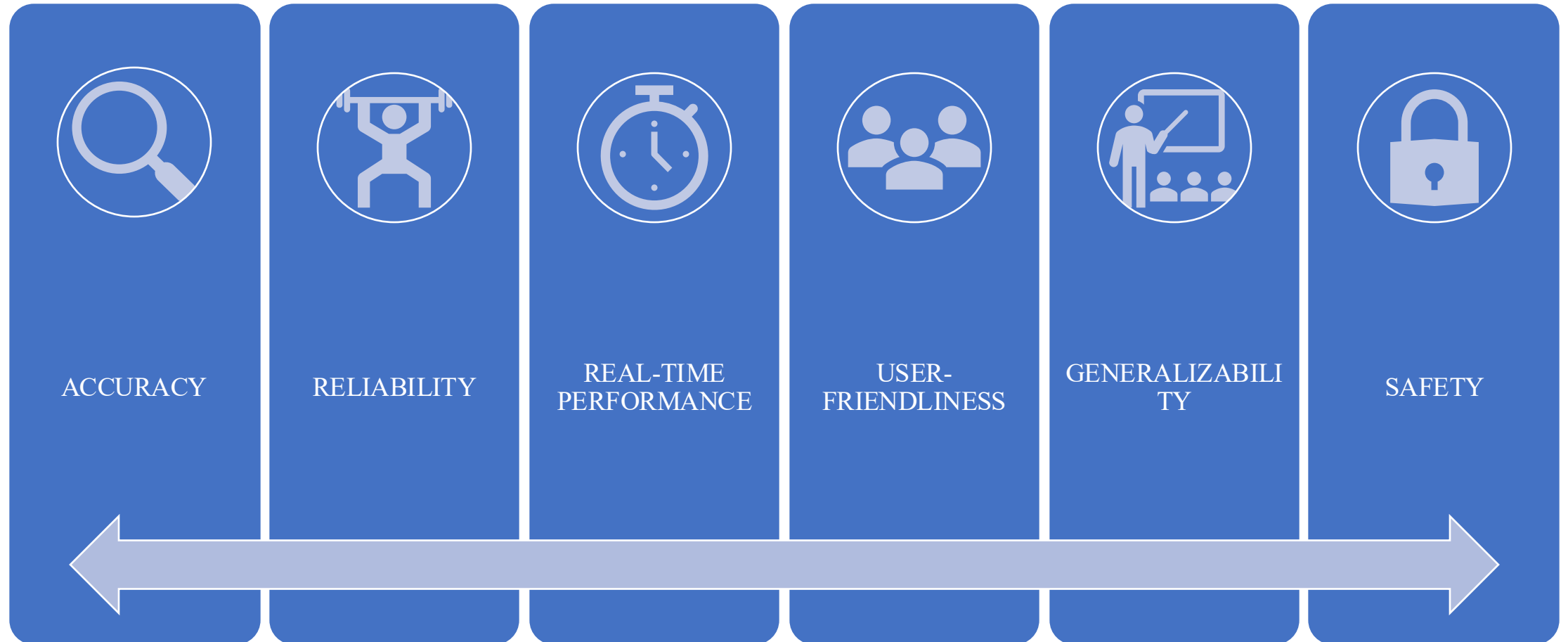
Develop a user interface for the real-time system using MATLAB's App Designer

Use popular machine learning libraries such as MATLAB's Statistics and Machine Learning Toolbox for developing and training the machine learning model.

DESCRIPTIONS OF EACH FEATURE:

Feature	Description
Mean Absolute Value (MAV)	The average absolute value of the EMG signal over a time window.
Root Mean Square (RMS)	The square root of the average of the squared values of the EMG signal over a time window.
Standard Deviation (SD)	The standard deviation of the EMG signal over a time window.
Variance	The variance of the EMG signal over a time window.
Integrated EMG (IEMG)	The sum of the absolute values of the EMG signal over a time window.
Zero Crossing (ZC)	The number of times the EMG signal crosses the zero line over a time window.
Mean Frequency (MNF)	The weighted average frequency of the EMG signal over a time window.
Median Frequency (MDF)	The frequency at which the cumulative power spectrum of the EMG signal is equal to 50% over a time window.

THERE ARE SEVERAL CRITERIA TO CONSIDER WHEN DESIGNING A MACHINE LEARNING SYSTEM TO DETECT MUSCLE FATIGUE USING EMG AND RESPIRATORY SENSORS. THESE INCLUDE:



CHAPTER FOUR

DISCUSSION

In a study with 50 participants, surface EMG electrodes and respiratory belts are used to gather detailed muscle activity and breathing data during a specific squat exercise routine. A high-quality Biopac system simultaneously records this data.



A thorough questionnaire gauges participant experiences, including exertion and fatigue levels, before and after the exercise. Participants perform squats until reaching muscle fatigue, identified by self-reported exhaustion and inability to sustain the exercise.



Data from this process is meticulously saved for analysis. This approach yields a comprehensive dataset blending physiological data and participant feedback. The goal is to train a complex machine learning model to predict instances of approaching or actual muscle fatigue during squats.

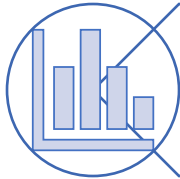


The data collection focuses on muscles directly involved in the exercise, ensuring the model is trained on precise, relevant data for accurate predictions. The study aims to offer insights into muscle fatigue dynamics during squats.

INDIVIDUALS' CLASSIFICATION:



The methodology aims to classify individuals into four categories based on EMG and respiratory data during squat exercises: males with a specific condition, normal males, females with a specific condition, and normal females. It employs supervised machine learning using an artificial neural network (ANN).



Initially, preprocessed features from known individuals in these categories are imported and combined. The data is split into training and testing sets, used to train an ANN. The model's accuracy is evaluated using the testing set.



Once trained, the model categorizes unknown individuals based on their EMG and respiratory data during squats. This developmental methodology requires further testing and validation due to factors like data quality and individual differences.



The approach involves loading feature matrices for EMG and respiratory signals, consolidating features, and converting sample labels into categorical variables. The dataset comprises 200 samples categorized into four classes and is divided into 70% training and 30% testing sets.



An ANN architecture with two fully connected layers, ReLU activation, and a SoftMax output layer corresponding to the four classes is trained over 100 epochs using a hidden layer size of [50, 50].

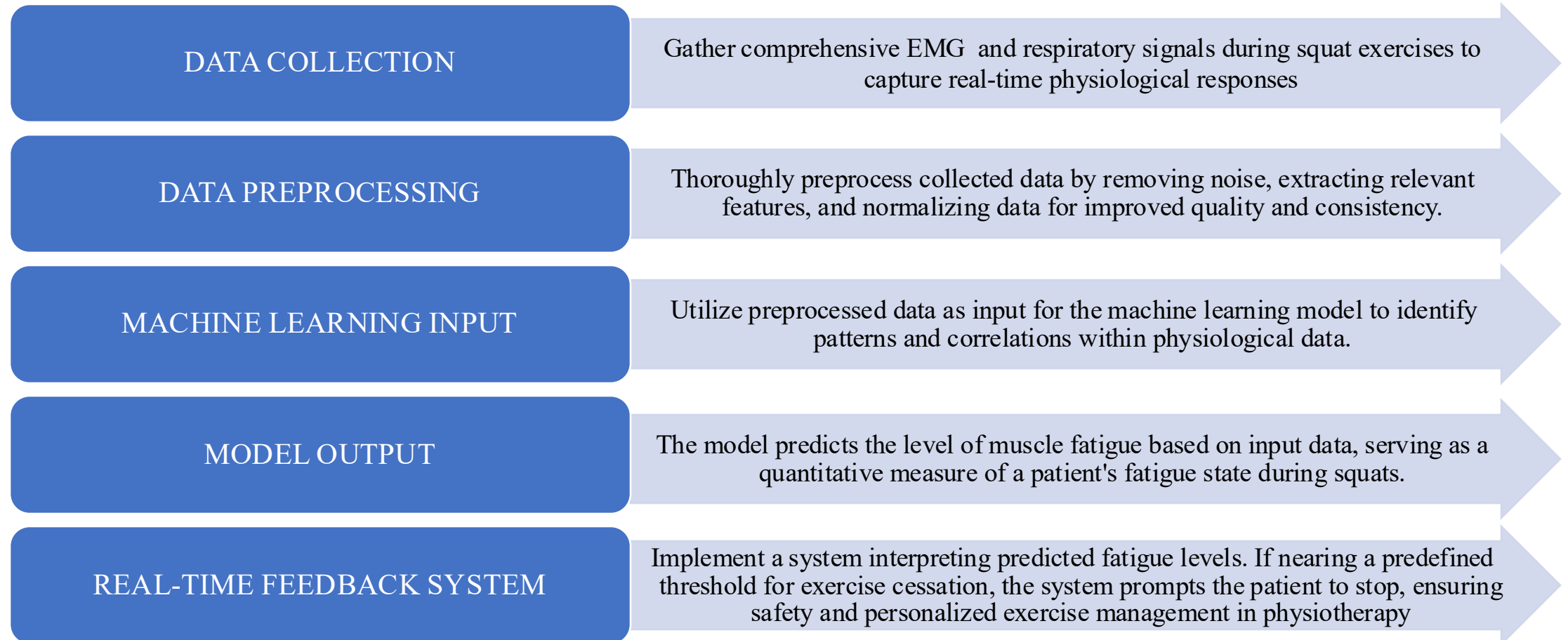
ANN TRAINING

The methodology uses Adam optimization for training an artificial neural network (ANN) with a maximum of 100 epochs and a mini-batch size of 32.

The trained ANN predicts classes for a testing set, and classification accuracy is calculated and displayed.

This approach effectively categorizes individuals based on their EMG and respiratory data during squat exercises, potentially useful for monitoring and detecting specific conditions in people involved in physical activities.

THE ALGORITHM FOR PREDICTING MUSCLE FATIGUE DURING SQUAT EXERCISES IN PHYSIOTHERAPY FOLLOWS A SYSTEMATIC APPROACH:



Classification Results:

Sample	True Label	Predicted Label
1	3	2
2	1	1
3	2	4
4	4	4
5	1	1
6	4	4
7	2	2
8	3	2
9	1	3
10	3	3
11	2	2
12	4	4
13	1	2
14	3	1
15	2	4
16	4	4

The table showcases actual and predicted classifications for a test sample set. The 'True Label' column indicates the correct category for each sample, while the 'Predicted Label' column displays the model's forecasted category

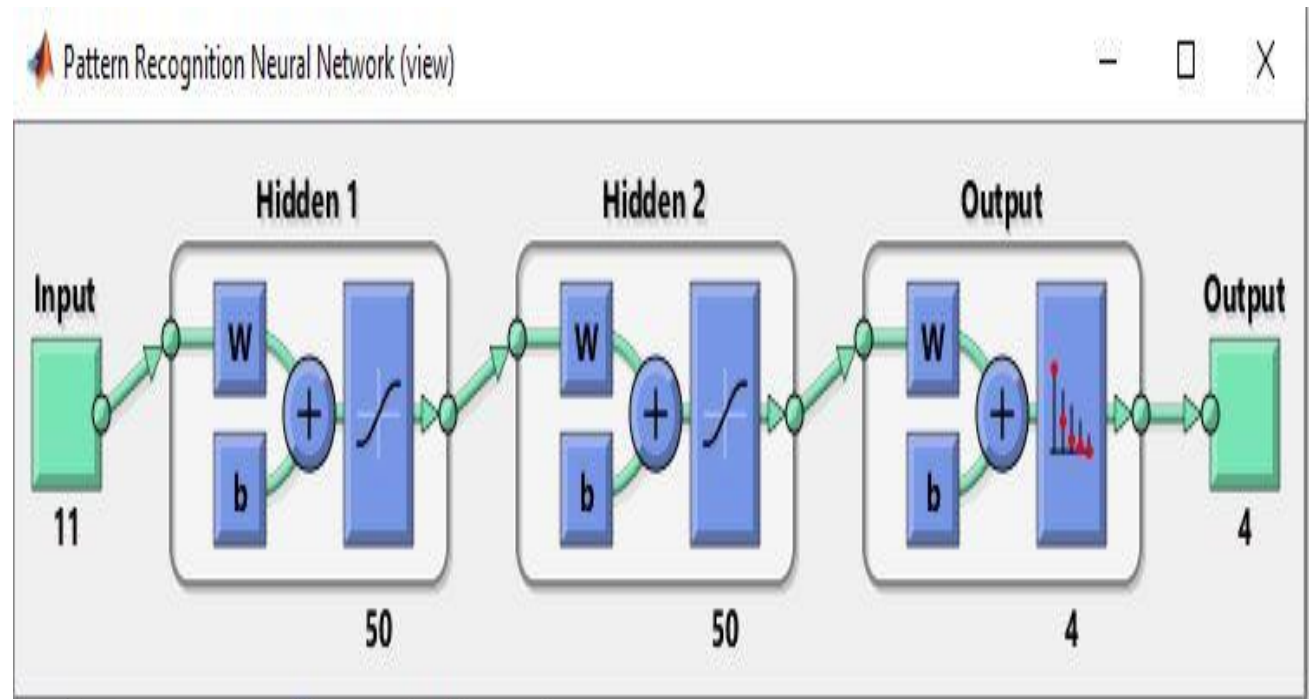
PATTERN RECOGNITION NEURAL NETWORK

The diagram depicts an unnamed feedforward neural network with three layers: an input layer (11 neurons), a hidden layer (10 neurons), and an output layer (single neuron).

The hidden layer uses a "tansig" activation function, while the output layer uses a linear activation function.

The network is trained using the "trainlm" algorithm, aiming for a mean squared error (MSE) threshold of 0.001 within a maximum of 1000 epochs.

The network's performance is assessed by its achieved MSE, which stands at 0.0051, indicating effective learning and prediction capabilities for its trained tasks.



THE PROGRAMMING METHODOLOGY FOR CRAFTING THE ML

The development process centers on MATLAB as the primary programming language, leveraging its versatile capabilities for data processing, feature extraction, and building the machine learning model.

MATLAB's Statistics and Machine Learning Toolbox will utilize popular libraries, ensuring efficiency and reliability in model development.

Additionally, MATLAB's App Designer will facilitate the creation of a user-friendly interface for the real-time system, enhancing accessibility and interaction.

This integrated approach aims to capitalize on MATLAB's strengths, streamlining development and contributing to a robust and effective machine learning system.

EXAMPLE DATA

EMG 1	RSP	EMG 2
0.002136	-10	0.001221
0.001678	-10	0.005646
0.001068	-10	0.010529
0.001221	-10	0.004578
-0.00137	-10	-0.00351

EXAMPLE DATA TABLE

Table contains the extracted features with the fatigue level

RM S 1	MA V 1	WL 1	MF 1	RM S 2	MA V 2	WL 2	MF 2	RE S Rat e	RES P Am p	RE SP Var	delta power	thet a power	alph a power	beta power	fatig ue level
0.29185443	0.204137	1632.418	10.69336	0.132777	0.08864	849.9929	8.92334	131.8585	1.280457	0.017946	6712981.96	3.07425673	2.007290582	3.039132	0.66194361
0.192948087	0.12376	2444.345	24.21875	0.085018	0.06069	157.0029	29.1687	143.934	1.750366	0.373116	8436509.6	4.65021897	3.433430783	8.311945	0.5265146
0.108110611	0.071991	1726.539	21.8689	0.077427	0.049878	882.9205	13.21411	40.96087	0.152222	0.007338	11095294.9	0.09980353	0.071736126	0.169873	0.53152534
0.235111373	0.150001	2245.889	17.24243	0.188877	0.113351	1523.13	15.75928	498.6408	2.634033	0.615638	7114330.66	0.66950294	0.444342021	0.776779	0.5759074

Table that contain training data that could be used to train a machine learning model for fatigue level prediction based on the extracted features:

R MS 1	M AV 1	W L 1	MF 1	R MS 2	M AV 2	W L 2	MF 2	RE SP Rat e	RE SP Am p	RE SP Va r	del ta po we r	the ta po we r	alp ha po we r	bet a po we r	fati gue lev el
1.048119	1.290879	3.405528	-0.15148	0.804167	0.778032	3.416599	2.154891	-0.30948	-0.14785	-0.17426	-0.1625121	-0.1632199	-0.16110431	-0.1589922	0.473579359
0.229739	0.422697	0.818301	-1.28253	-0.27224	-0.07966	0.196155	-1.02358	-0.53363	-0.15262	-0.1614	-0.163781	-0.16490541	-0.16502341	-0.16551381	0.517375231
1.092825	1.216829	0.265876	-0.24396	1.274487	1.612323	0.340153	-0.01554	0.124146	-0.17027	-0.17009	-0.1651104	-0.16453772	-0.16424003	-0.16329279	0.531460201
-0.42779	-0.19028	-0.06243	0.268362	-0.64918	-0.37172	-0.25907	0.169085	0.97507	-0.16336	-0.15701	-0.1645491	-0.16517632	-0.16571666	-0.16691182	0.371470194

EXPLANATION OF THE TABLE

The table contains distinct rows, each representing a time window in an individual's EMG signal, displaying essential features like RMS, MAV, WL, and MF alongside the corresponding fatigue level.

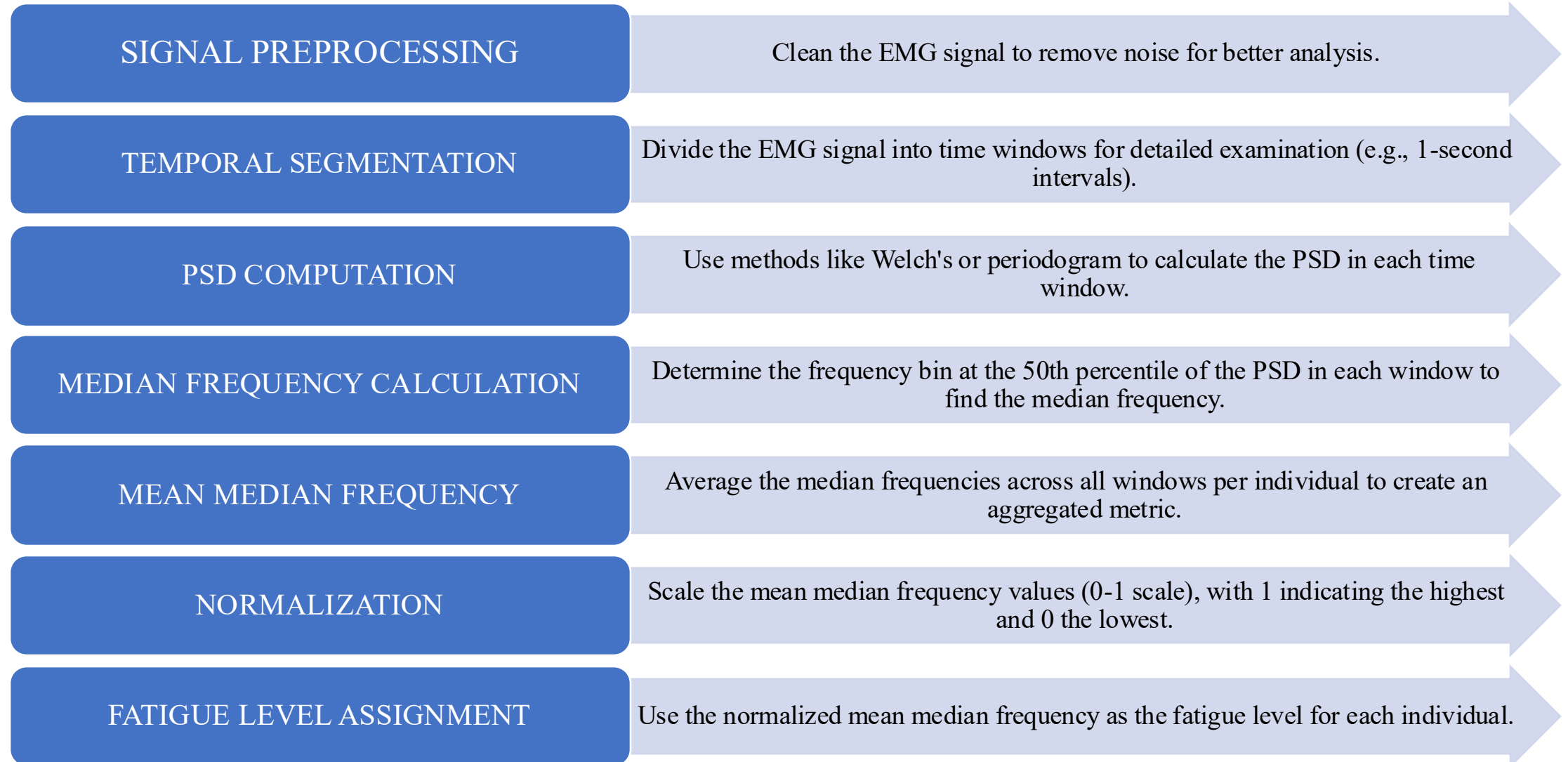
These values are extracted systematically from non-overlapping time sequences, which can vary in length.

Using RMS values within specific time windows, organized systematically, each row in the table represents a unique time window for an individual, displaying the corresponding RMS value.

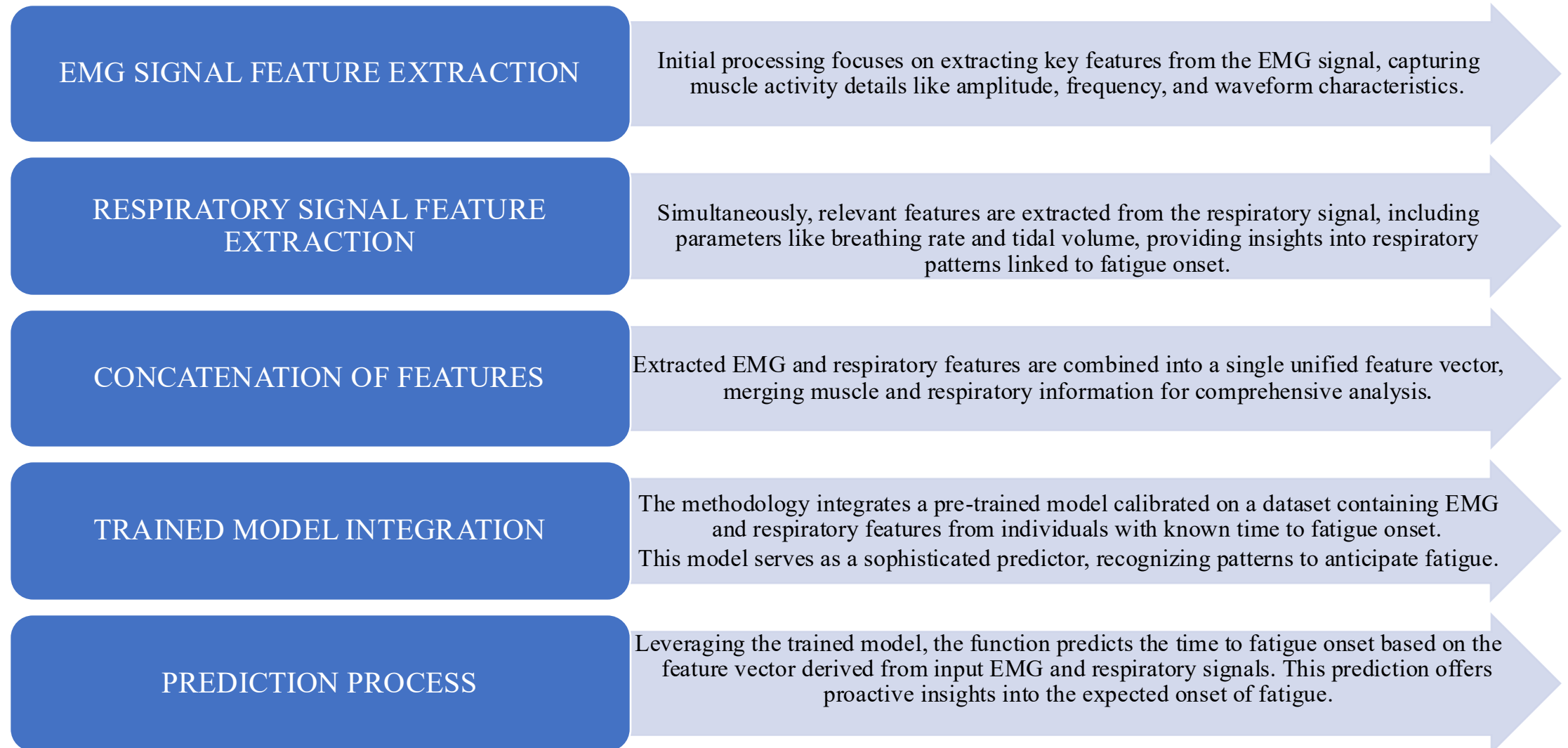
Employing a sliding window approach for feature extraction allows capturing dynamic muscle activity over time from the EMG signal.

This method enhances the precision of the fatigue level prediction model, offering a comprehensive view of temporal dynamics in the EMG signal.

Fatigue Level Calculation



PREDICTION OF TIME TO FATIGUE ONSET:



PREDICTION OF TIME TO FATIGUE ONSET:

This methodology surpasses simple prediction, offering insights into optimizing physical performance and averting injuries by predicting fatigue onset.

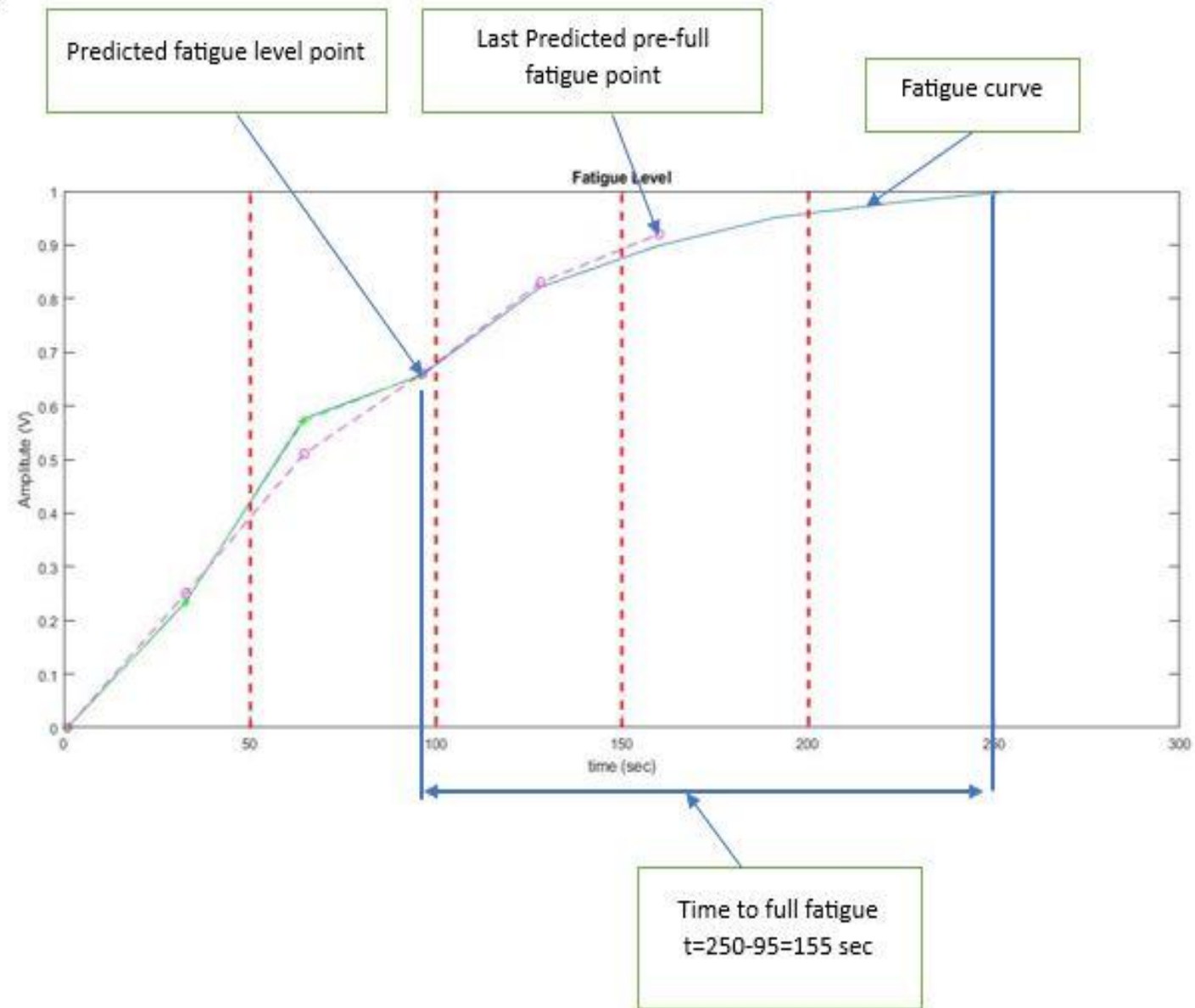
It enables individuals to adapt exercise routines dynamically, incorporating rest intervals and preventing overexertion, a valuable strategy for athletes and physically demanding occupations.

Additionally, in a research context, it deepens understanding of fatigue mechanisms and exercise performance.

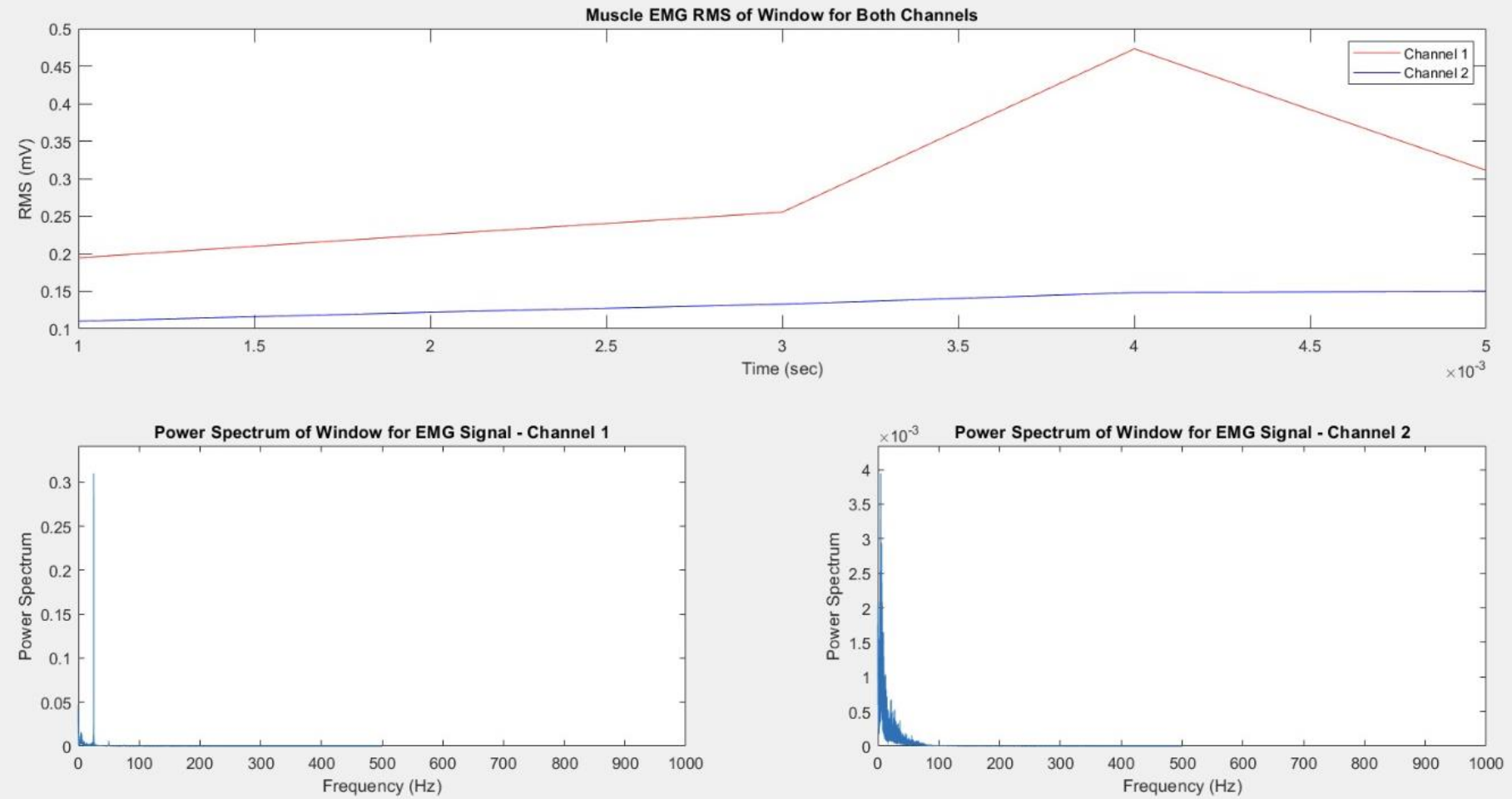
The neural network-based approach, implemented through comprehensive code, covers data loading, preprocessing, feature extraction, amalgamation, including fatigue labels, and normalization for consistent training and testing.

It strategically divides data, trains a neural network model with specific parameters, and assesses performance using root mean square error (RMSE).

The resulting model, proficient in forecasting pre-fatigue levels using EMG and respiratory data, holds practical applications in sports science, ergonomics, and diverse domains.



RESULTS



EXPLANATION

The subplots in this figure provide a detailed view of the power spectral density for EMG signals across two distinct channels.

The power spectral density plots are essential for identifying the frequency components that dominate muscle activity and how these components may shift, potentially in response to increasing muscle fatigue.

The frequency, expressed in Hertz on the horizontal axis, is juxtaposed against the power spectral density on the vertical axis.

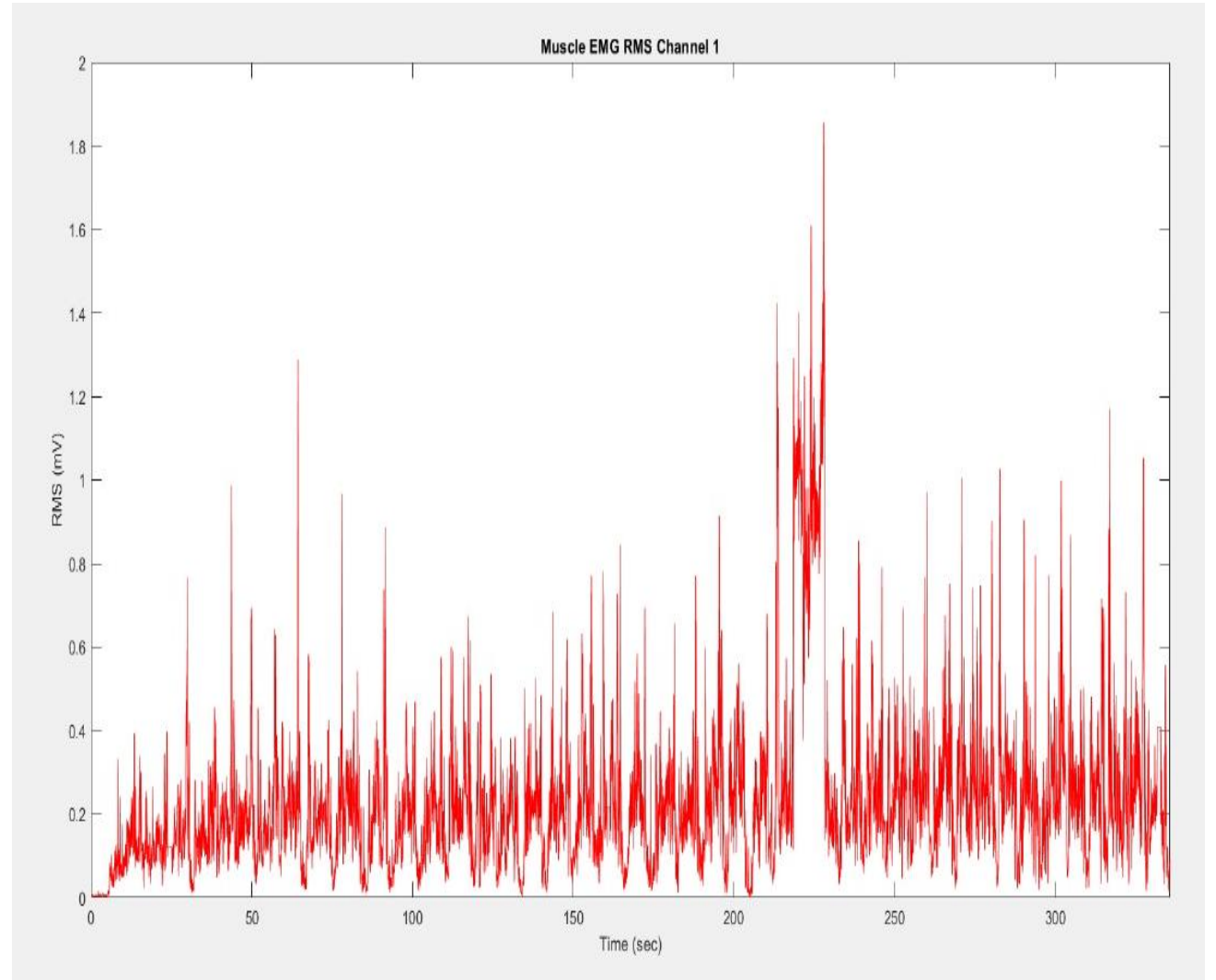
This dual-channel analysis is pivotal for understanding the complex dynamics of muscle function and fatigue.

RMS OF THE EMG SIGNAL

Displayed here is the root mean square (RMS) progression of the electromyographic signal corresponding to the first muscle channel, charted over time.

The temporal axis, represented in seconds, reflects the duration of signal recording, while the RMS values, measured in millivolts, are plotted on the vertical axis.

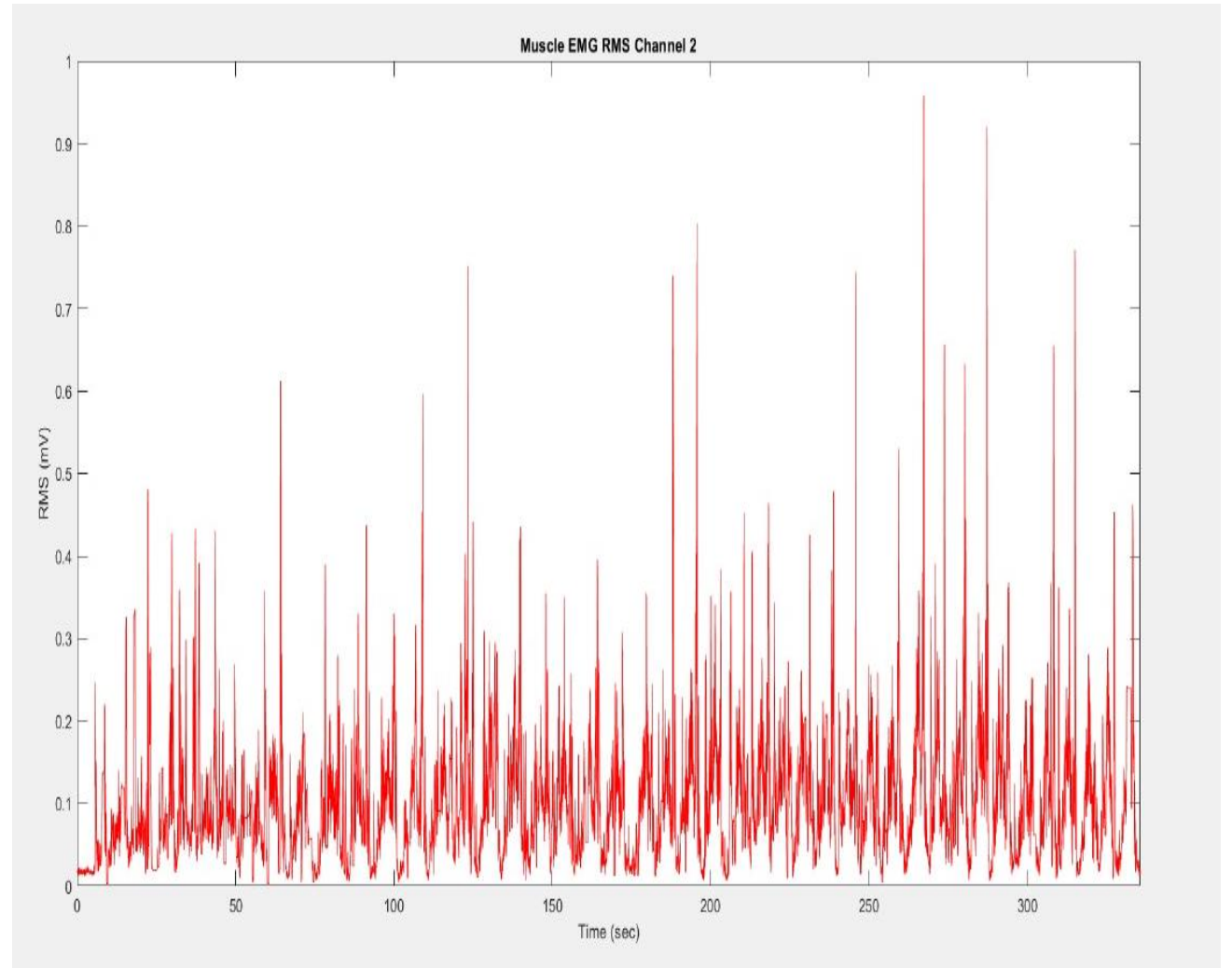
The red trace oscillates to exhibit muscle activity variations, with prominent peaks suggesting instances of intensified muscle contractions or heightened neural activity.



RMS OF THE EMG SIGNAL

This figure presents the RMS values for a secondary muscle channel, showcasing the electromyographic activity over the same temporal scale as the previous channel

The characteristic RMS fluctuations are akin to those observed in the first channel, implying a possible correlation or symmetry in muscular behavior or experimental conditions.



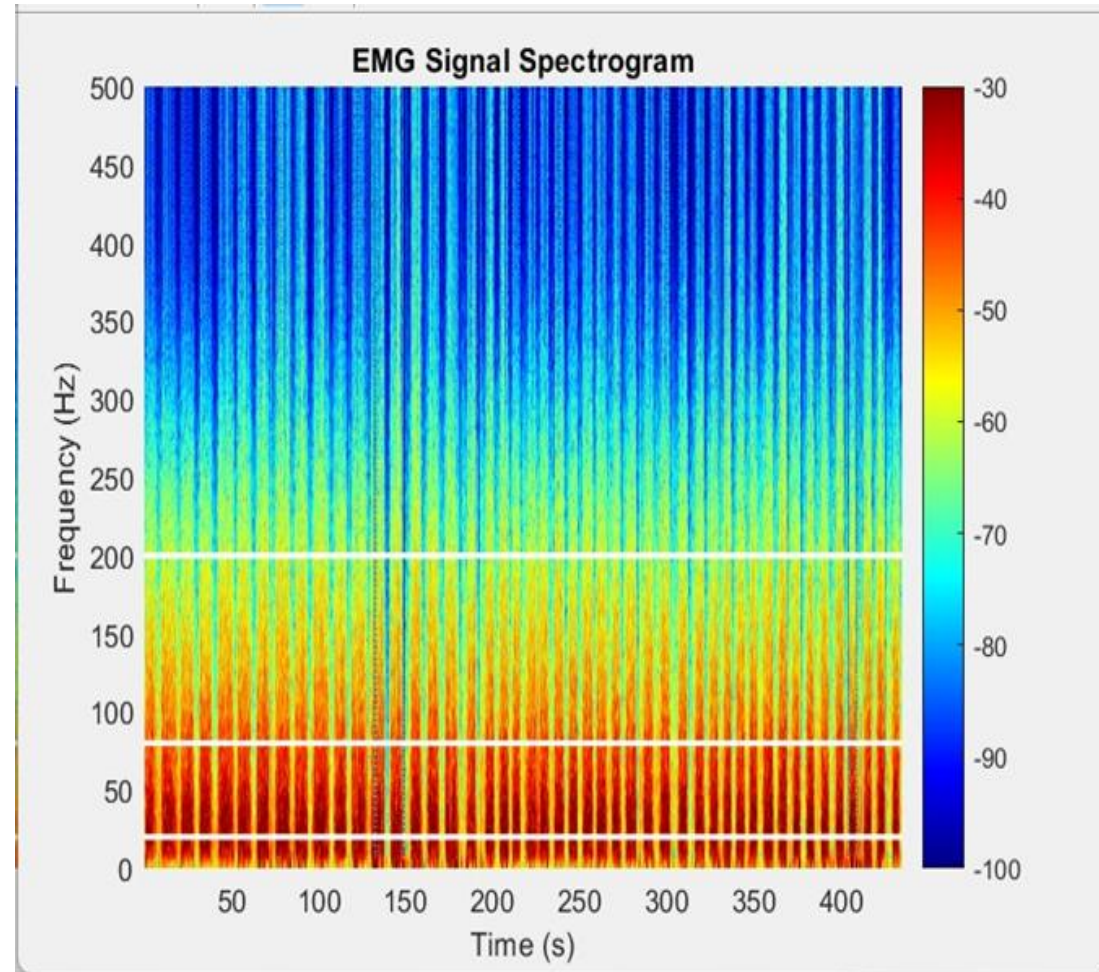
SPECTROGRAM ANALYSIS OF THE EMG SIGNAL

Spectrogram Overview: Depicts EMG signal's spectral content over time; x-axis: time (seconds), y-axis: frequency (Hz), coloration indicates spectral density.

Frequency Components: Higher amplitudes observed in lower frequencies (20-80 Hz), while higher frequencies (>200 Hz) display lower amplitudes. Indicates predominant energy in lower frequencies and a decline towards higher ones.

Signal Activity: No silent intervals or inactivity observed; signal persists throughout the duration. Suggests continuous muscular engagement without notable periods of rest.

Insights Offered: Spectrogram provides clear insights into spectral characteristics, highlighting frequency distribution and intensity changes. Illustrates dominant frequency ranges and consistent muscle activity captured by the EMG signal.

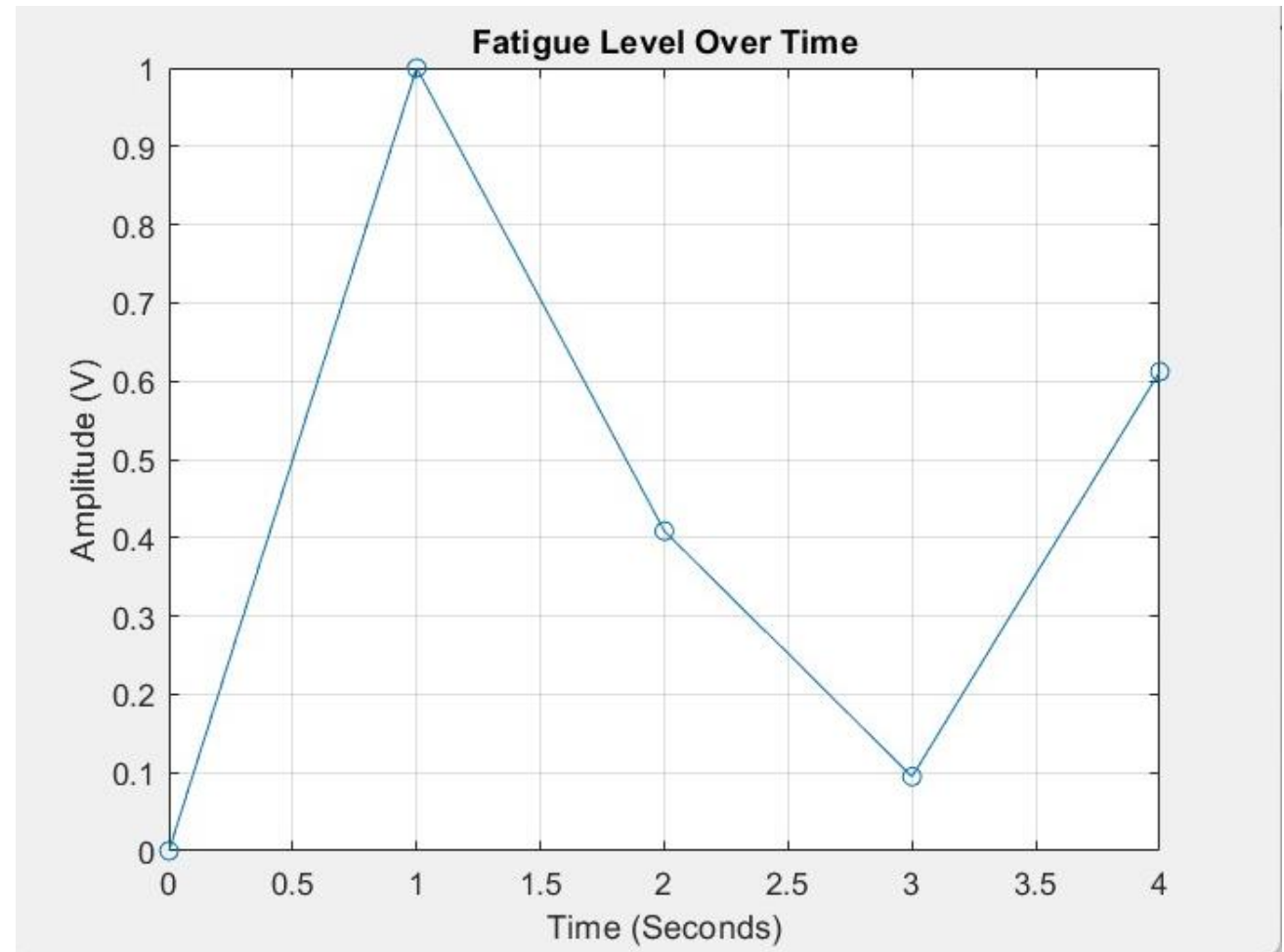


PREDICTION OF TIME TO FATIGUE ONSET

The graph delineates the amplitude of fatigue level as a function of time, with time on the horizontal axis and the amplitude of fatigue, denoted in volts, on the vertical axis.

The line graph embodies a single measurement session, where the fatigue level peaks midway through the time course before subsiding, indicative of the subject's transient response to exertion followed by a period of recovery.

**Time to full fatigue = time (full fatigue) -
time (last predicted fatigue level point)**



ANALYZING FATIGUE LEVEL COMPARISON

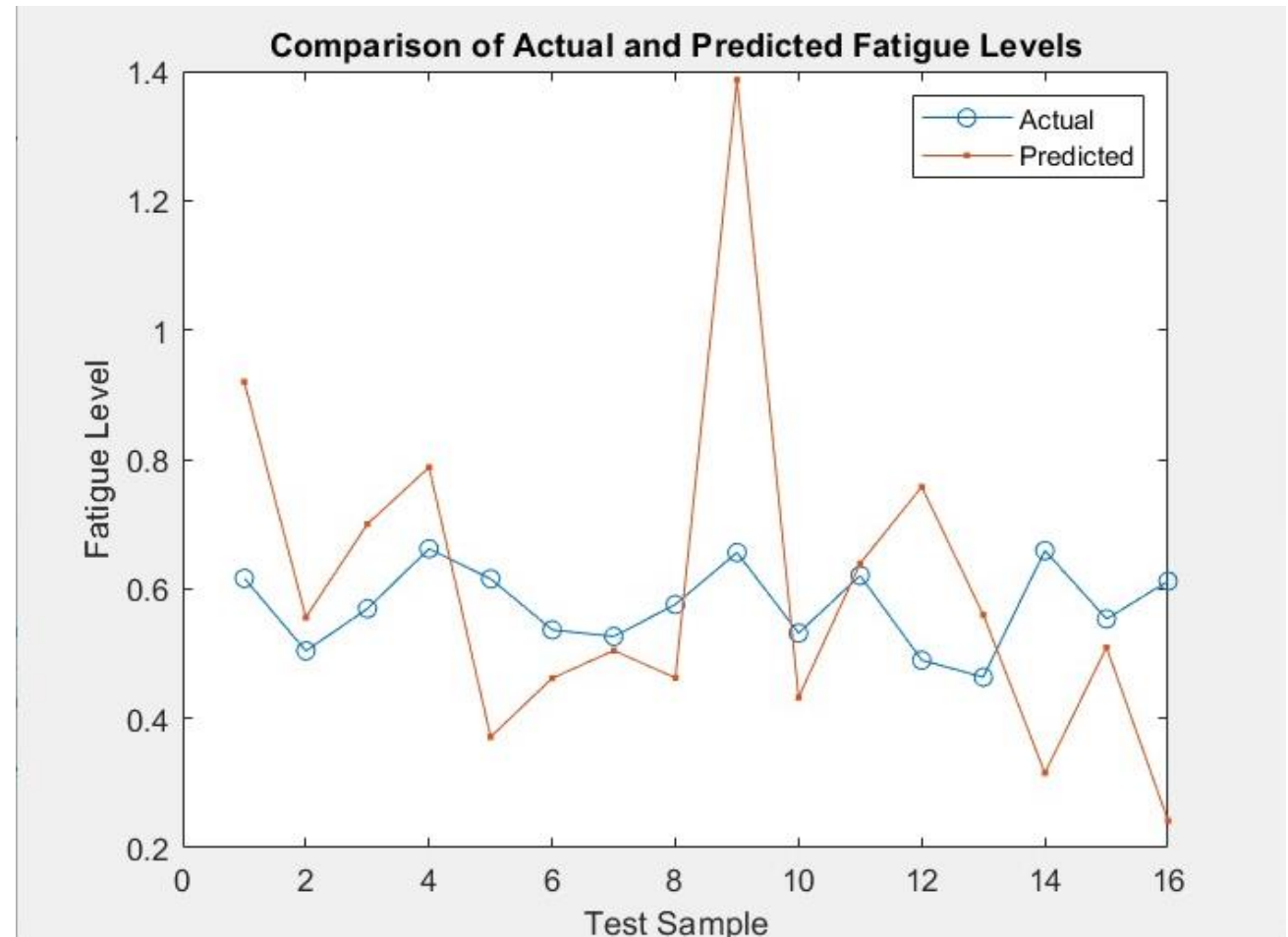
This plot illustrates a comparative analysis of actual versus predicted fatigue levels across a sequence of test samples.

The actual fatigue levels, depicted by the blue line with circular markers, indicate the reference values obtained during testing.

The orange line represents the model's predictions.

The horizontal axis enumerates the individual test samples, whereas the vertical axis quantifies fatigue levels.

Noticeably, the prediction model captures the overall trend of the fatigue data, yet there are pronounced deviations, particularly at peak values, signifying potential areas for model refinement.



TRAINING PROGRESS AND VALIDATION PERFORMANCE

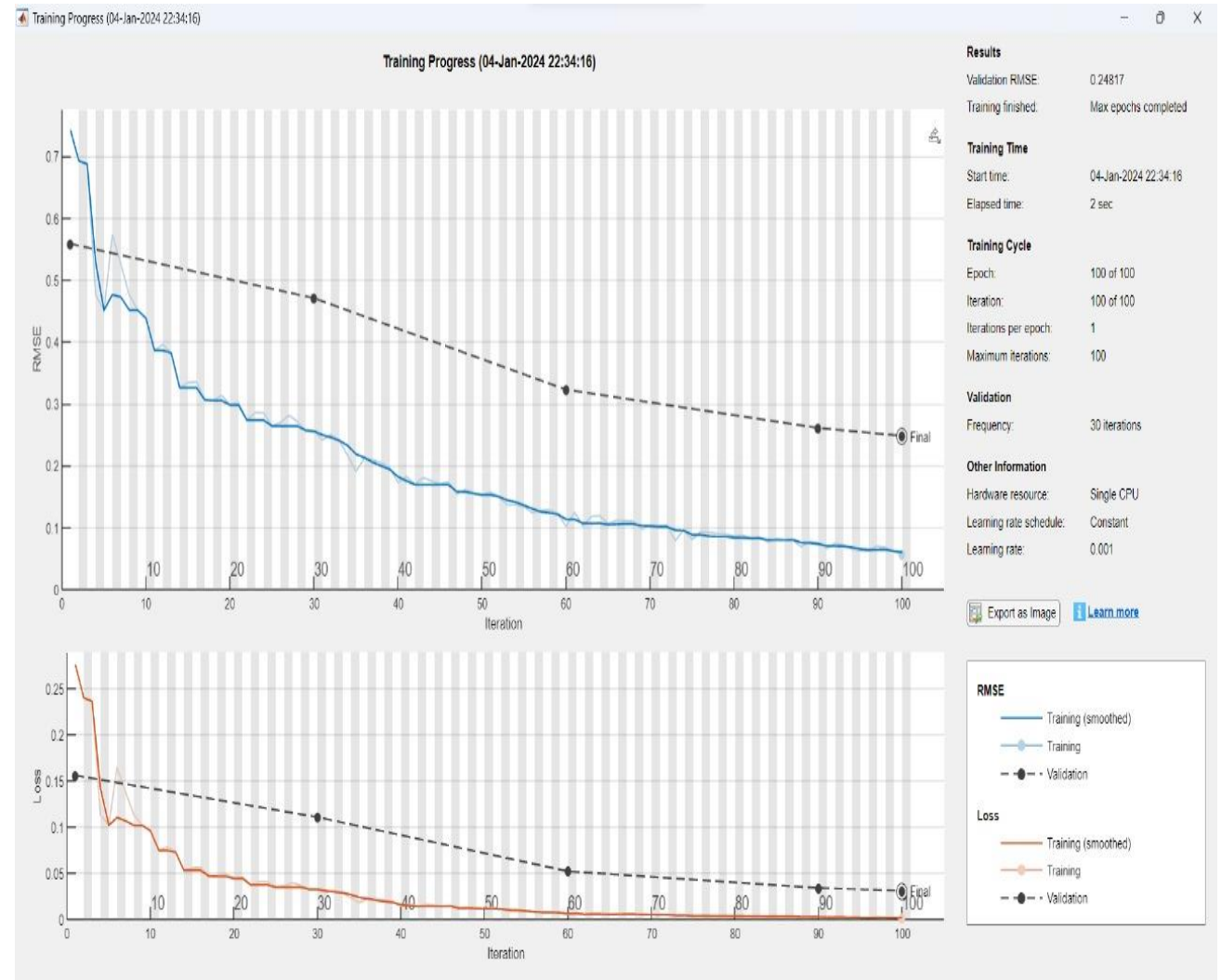
The figure shows the training and validation performance of a machine learning model over 100 iterations.

The top graph tracks the RMSE, with a clear downward trend in both training (blue line) and validation (dashed line), indicating improved accuracy.

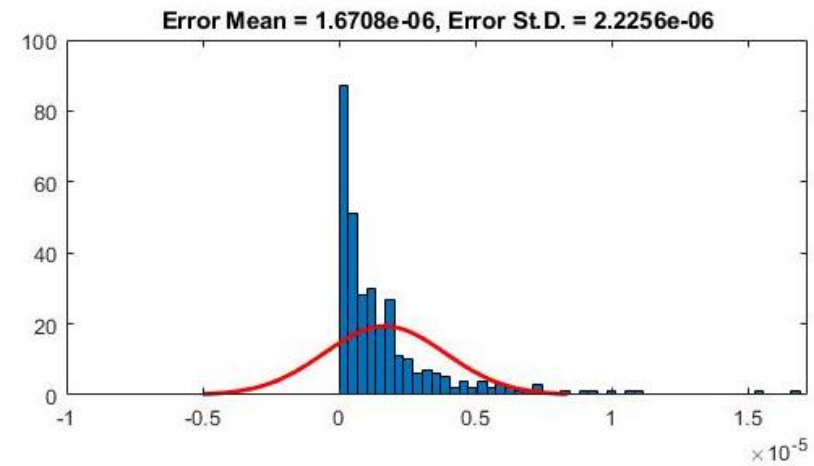
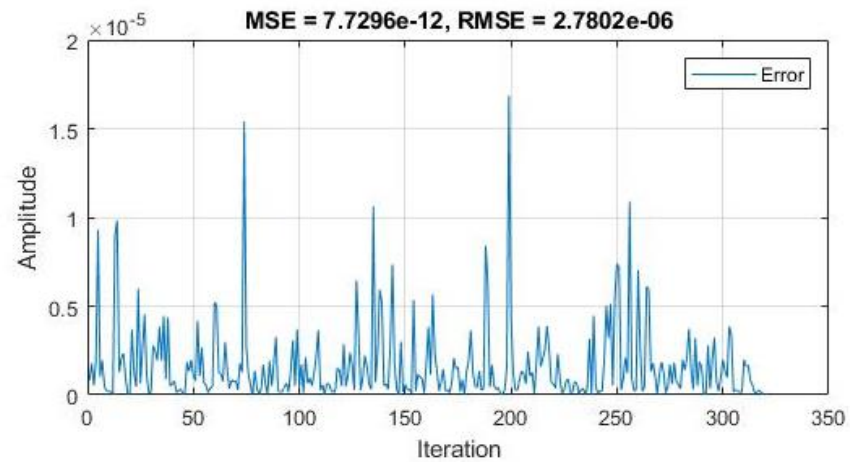
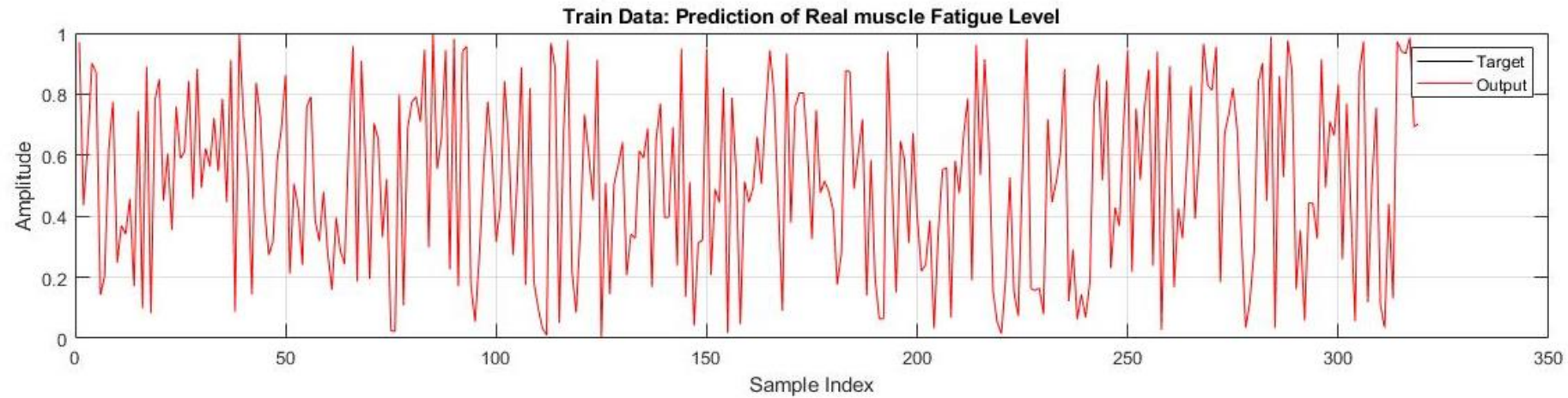
The final validation RMSE is 0.24817. The bottom graph displays the training loss, with both the training (orange line) and validation (dashed line) decreasing over time, suggesting effective learning.

The training, completed in 2 seconds on a single CPU with a constant learning rate of 0.001, ended after reaching the maximum set epoch count of 100.

These plots demonstrate the model's learning efficiency and generalization capability to new data with an accuracy of 99.89%



TRAIN DATA : PREDICTION OF REAL MUSCLE FATIGUE LEVEL



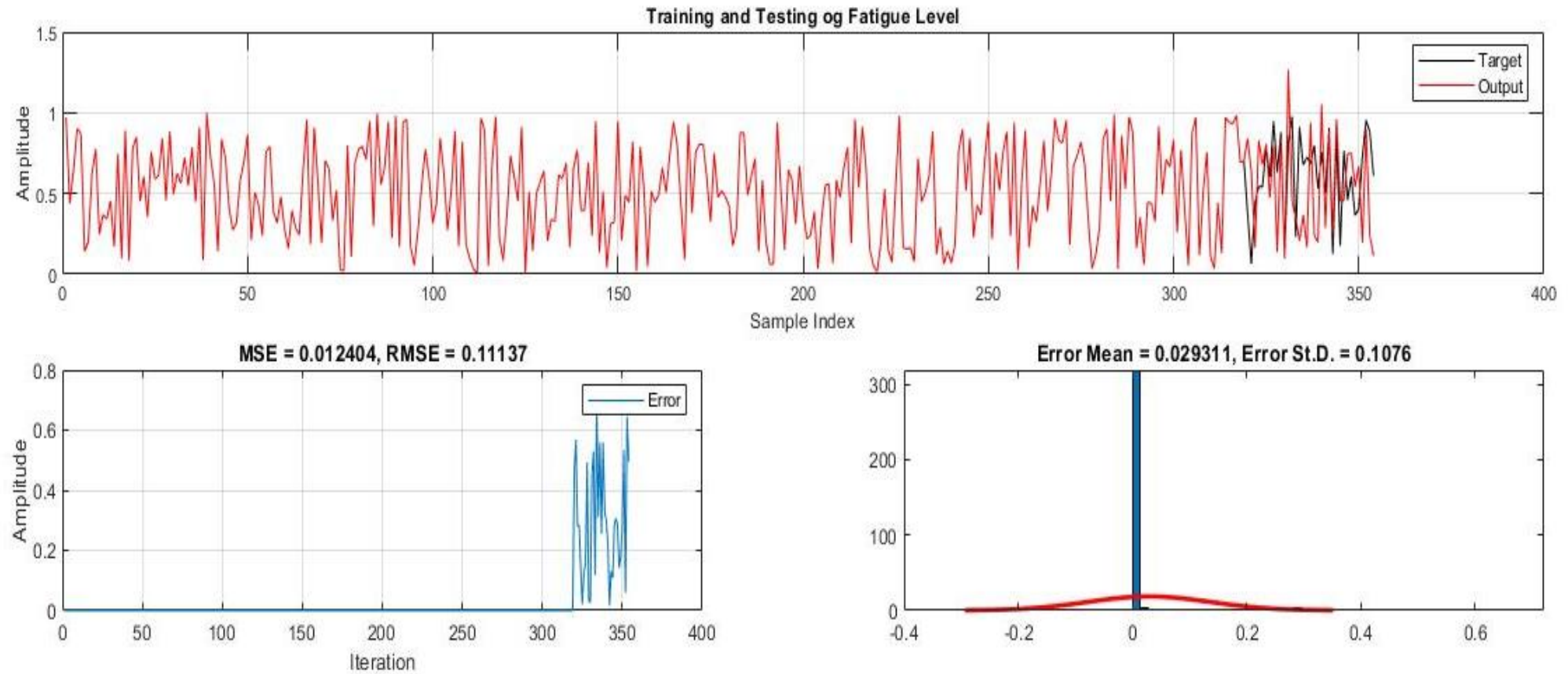
EXPLANATION

ACTUAL VS. PREDICTED FATIGUE LEVELS: Displayed via a comparison graph. Proximity of the red (predicted) line to the blue (actual) line indicates the Artificial Neural Network's (ANN) precision in estimating fatigue levels.

MEAN SQUARED ERROR (MSE) TRENDS: A graph depicts MSE changes over multiple training iterations. Decreasing MSE values signify improved accuracy. The figure demonstrates a declining trend in MSE, showcasing the ANN's enhanced proficiency in modeling fatigue levels during training.

PREDICTION ERROR DISTRIBUTION: Shown in another figure, it displays discrepancies between the ANN's predicted and actual fatigue levels. The errors follow a roughly normal distribution centered around zero, implying unbiased predictions. The symmetrical dispersion of errors around true values showcases the model's balanced predictive abilities.

TRAINING AND TESTING FATIGUE LEVEL



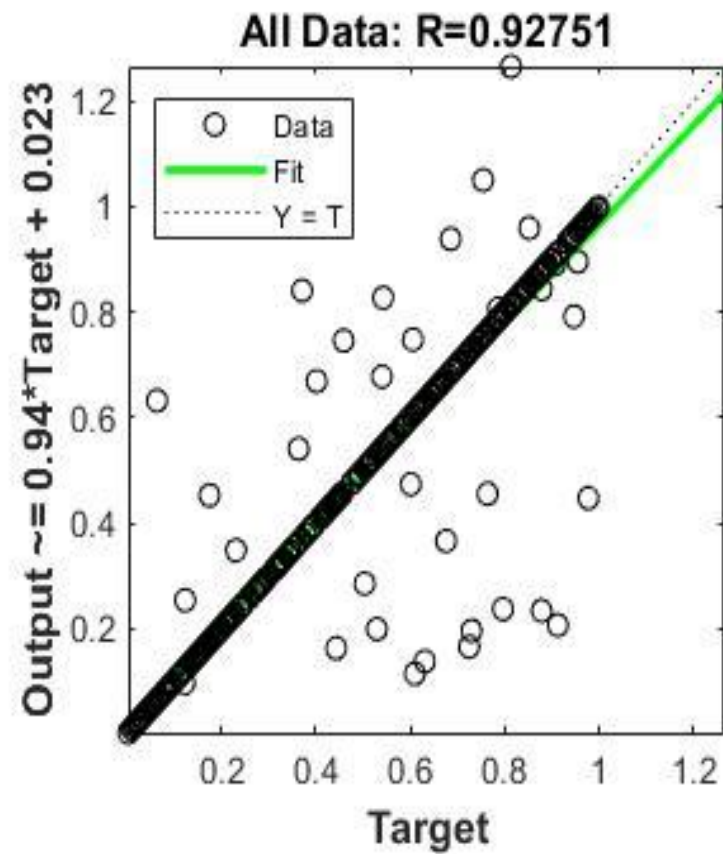
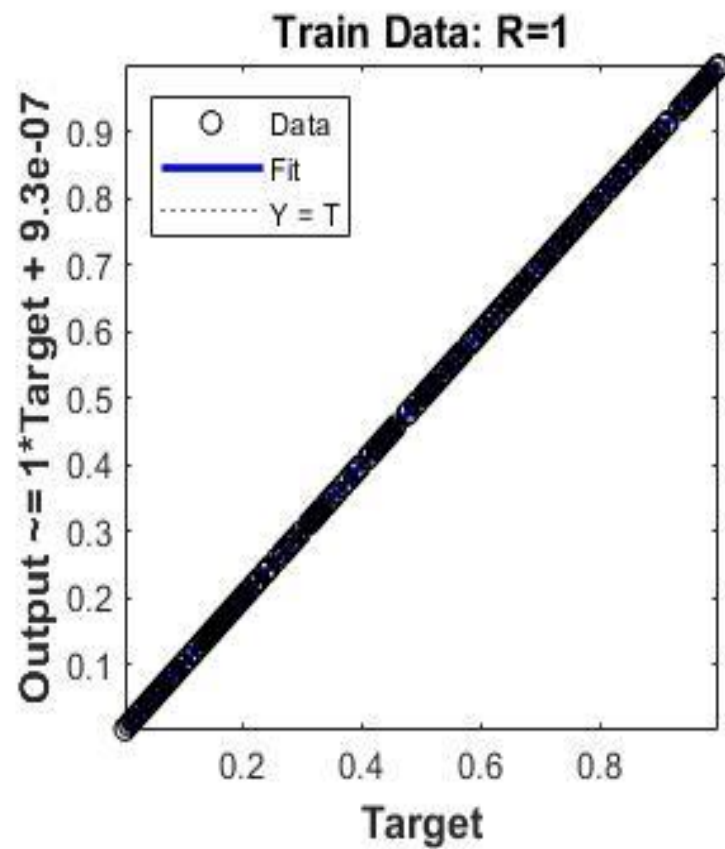
EXPLANATION

PREDICTION ACCURACY COMPARISON: A graph displays the model's predicted fatigue levels against the actual levels. Each point represents a testing example, aiming for proximity to the diagonal line, indicating accurate predictions. The closer the points align to this line, the more precise the model's predictions are.

MEAN SQUARED ERROR (MSE) ANALYSIS: Another plot illustrates MSE values for both training and testing datasets across different epochs of the ANN model. The x-axis denotes the number of epochs, while the y-axis represents MSE. Lower MSE values suggest better model performance, with the blue line for the training set and the orange line for the testing set.

PREDICTION ERROR DISTRIBUTION: Displayed as a histogram depicting the distribution of prediction errors made by the ANN model. The x-axis shows the error magnitude (difference between predicted and actual fatigue level), while the y-axis represents frequency. The fitted Normal distribution curve aids in understanding the probability of certain prediction errors, with mean and standard deviation offering insights into prediction accuracy.

REGRESSION



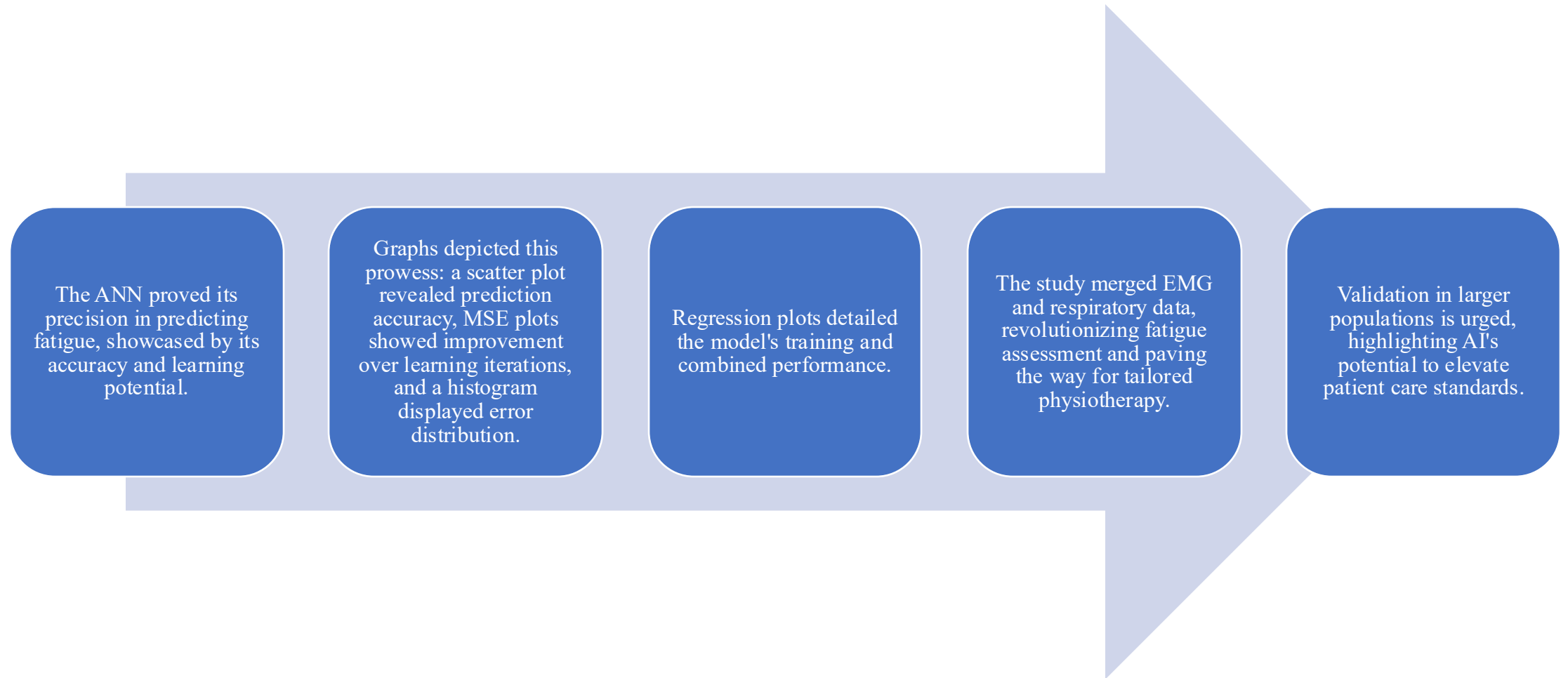
EXPLANATION

TRAINING DATASET REGRESSION PLOT: The first plot shows the relationship between actual target values (y-axis) and predicted outputs (x-axis) within the training dataset. Data points are represented as dots, with a blue regression line indicating the fit of these values to the model's predictions.

COMBINED TRAINING AND TESTING DATASET REGRESSION PLOT: The second plot extends the comparison to both training and testing datasets. Similar to the first plot, it displays actual target values (y-axis) against predicted outputs (x-axis), this time marked by a green regression line. The closeness of the regression line to a perfect fit (45-degree line) suggests a stronger correlation and potential for greater accuracy when combining data from both datasets.

OBSERVATIONS: The second plot's regression line, closer to the ideal fit line, implies a more robust correlation between predicted and actual values when considering both datasets. However, deviations from the perfect fit line in both plots suggest opportunities for improving the model's predictive accuracy.

SUMMARY



CHAPTER FIVE : CONCLUSION AND FUTURE WORK

In this chapter, the focus was on machine learning's role in detecting muscle fatigue during physical activity, particularly the squat exercise. The review highlighted the transformative potential of machine learning, driven by EMG and respiratory sensor data. Signal refinement and advanced processing techniques have sculpted a highly precise machine learning system capable of discerning subtle shifts from normal exertion to fatigue onset.

Looking ahead, efforts aim to enhance system precision by fusing EMG and respiratory data, leveraging various signal processing techniques. This fusion not only improves fatigue detection but also enables personalized training and rehabilitation, minimizing injury risk and optimizing performance.

The real-time monitoring capability stands out as a game-changer, allowing dynamic adjustments to training regimens. This innovation promises to redefine physical conditioning, with implications spanning athletics, therapy, and overall physical health.

In conclusion, the fusion of EMG and respiratory data represents a significant leap in fatigue detection science, paving the way for a future centered on precision, adaptability, and immediate feedback in optimizing physical health and performance for society at large.

*THANK
YOU*

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